

Program : Diploma in Cloud Computing and Big Data	
Course Code : 5273C	Course Title: Data Mining Practices
Semester : 5	Credits: 4
Course Category: Programme Elective course	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To impart data mining concepts
- To impart data pre-processing techniques and their importance
- To impart data visualization methods
- To provide cluster analysis techniques.
- To provide various data mining approaches.
- To provide concept of outliers
- To provide insight into Applications of data mining.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basics in Python	1008	Introduction to IT systems Lab	1
Understand the concept of Database systems	3272	Database System Concepts	3

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Describe the concept of Data Mining and its applications.	12	Understanding

CO2	Explain the preprocessing of data and apply mining techniques	15	Understanding
CO3	Summarize data warehouse and mining techniques.	19	Understanding
CO4	Illustrate case studies of data mining applications	12	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the concept of Data Mining and its applications.		
M1.01	Describe Data Mining	3	Understanding
M1.02	Illustrate the kind of data that can be mined.	3	Understanding
M1.03	Explain the technologies used in data mining	3	Understanding
M1.04	Outline the issues in Data Mining	3	Understanding
Contents: Data mining, Types of data that can be mined, Patterns Mining, Kinds of patterns, Data mining Techniques, Applications of data mining, Issues in Data Mining			
CO2	Explain the preprocessing of data and apply mining techniques		

M2.01	Describe Data Cleaning	5	Understanding
M2.02	Explain Data Integration	4	Understanding
M2.03	Illustrate Data Reduction	4	Understanding
M2.04	Differentiate Data Transformation and Data Discretization	2	Understanding
	Series Test – I	1	
Contents: Data Cleaning-Missing Values, Noisy Data, Data Cleaning as a Process; Data Integration- Entity Identification Problem, Redundancy and Correlation Analysis, Tuple Duplication, Data Value Conflict Detection and Resolution; Data Reduction-Overview of Data Reduction Strategies, Wavelet Transforms, Principal Components Analysis, Regression and Log-Linear Models: Parametric Data Reduction, Histograms, Clustering, Sampling; Data Transformation and Data Discretization-Data Transformation Strategies Overview, Data Transformation by Normalization, Discretization by Binning, Discretization by Histogram Analysis, Discretization by Cluster, Decision Tree, and Correlation Analyses, Concept Hierarchy Generation for Nominal Data			
CO3	Summarize data warehouse and mining techniques.		
M3.01	Describe Data warehouse concepts	4	Understanding
M3.02	Summarize Data Visualization techniques	5	Understanding
M3.03	Explain various Clustering Algorithms	5	Understanding
M3.04	Outline Outliers and Outlier detection methods	5	Understanding
Contents : Data warehouse, Compare Database systems with Data warehouse, Multi-layered architecture of data warehouse, Extraction, Transformation and Loading, Meta data. Data Visualization- Data Objects and attribute types, Visualization techniques- pixel oriented, geometric projection, icon based, and hierarchical Clustering- Cluster Analysis, Partitioning Methods k-Means: A Centroid-Based Technique, Hierarchical Methods- Agglomerative versus Divisive Hierarchical Clustering, Distance Measures in Algorithmic Methods. Outlier Detection -Outliers, Types of Outliers , Challenges of Outlier Detection Supervised, Semi-Supervised, and Unsupervised Methods, Statistical Methods, Proximity-Based Methods, and Clustering-Based Methods, Applications			
CO4	Illustrate case studies of data mining applications		

M4.01	Classify different types of web data mining.	3	Understanding
M4.02	Explain Web log structure and issues regarding web logs	3	Understanding
M4.03	Describe Spatial Data Mining, Temporal Mining, And Multimedia Mining.	3	Understanding
M4.04	Summarize the Applications of Distributed and Parallel Data Mining.	3	Understanding
	Series Test – II	1	

Contents : Introduction to Web Mining, Web content mining, Web usage mining, Web Structure mining, Web log structure and issues regarding web logs.

Spatial Data Mining, Temporal Mining, And Multimedia Mining, Applications of Distributed and parallel Data Mining

Text / Reference

T/R	Book Title/Author
T1	J. Han, M. Kamber, “Data Mining Concepts and Techniques”, Morgan Kaufmann.
T2	M. Kantardzic, “Data mining: Concepts, models, methods and algorithms, John Wiley & Sons Inc.
R1	M. Dunham, “Data Mining: Introductory and Advanced Topics”, Pearson Education

Online Resources

Sl.No	Website Link
1	https://www.javatpoint.com/data-mining
2	https://www.talend.com/resources/what-is-data-mining/
3	https://www.ibm.com/cloud/learn/data-mining